

# Embedded Systems Design

## Chapter 1: Introduction to Embedded Systems

What is an Embedded System?

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Components of Microcontrollers

System-On- Chip and its examples

Introduction to embedded processor

# Introduction to Embedded System

What is Embedded System?

An embedded system is a dedicated computer system designed to perform one specific function or a set of related functions within a larger system. It consists of hardware and software integrated together, usually based on a microcontroller or microprocessor, and operates with real-time constraints.

An Electronic/Electro mechanical system which is designed to perform a specific function and is a combination of both hardware and firmware (

## Embedded system

**Hardware**

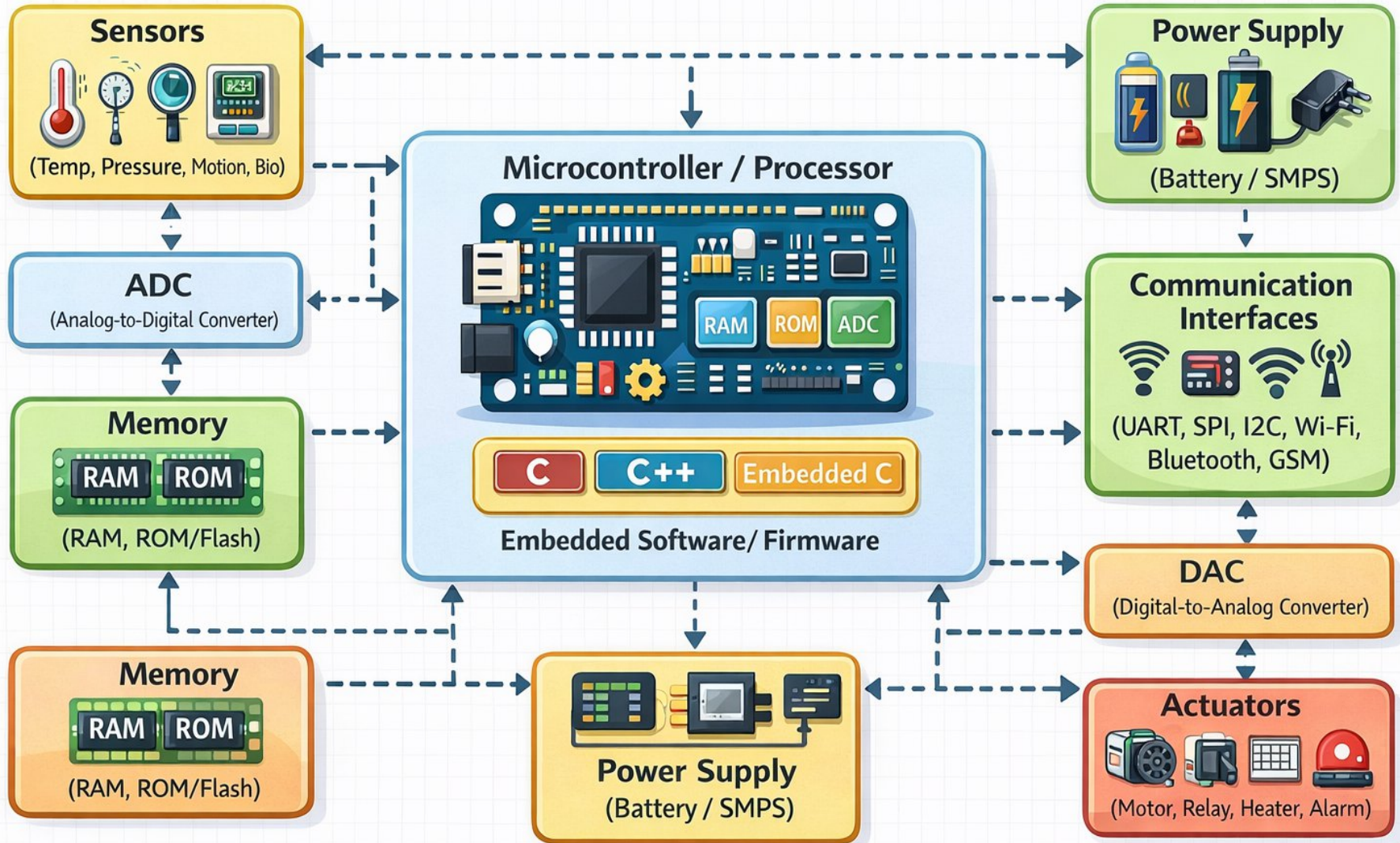


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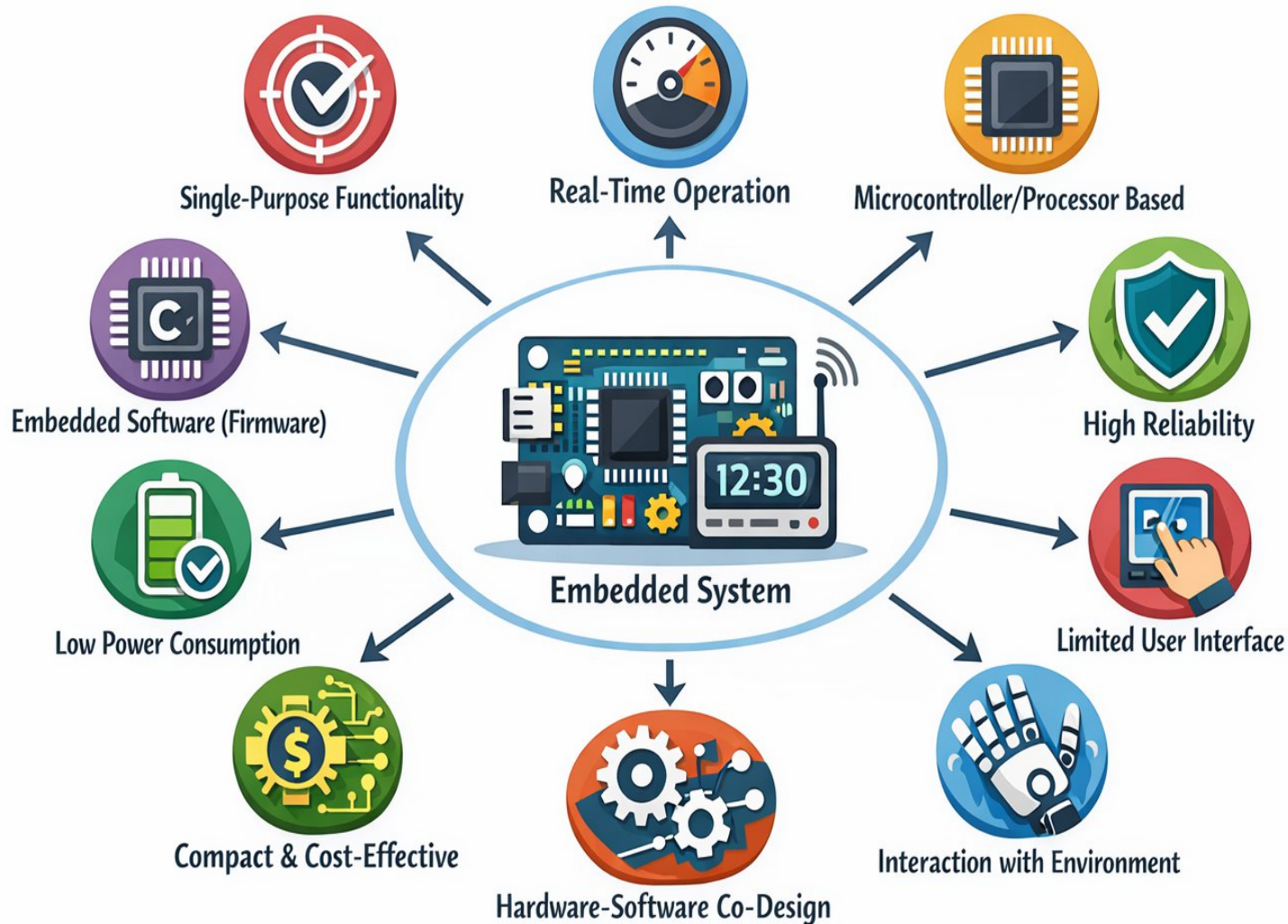
**Software**



# Components of an Embedded System



# Characteristics of Embedded Systems





## Characteristics of Embedded Systems

### 1. Single-purpose Functionality

Designed to perform a specific task (e.g., temperature control, heart rate monitoring).

### 2. Real-Time Operation

Must respond to inputs within a defined time constraint.

- 1. Hard real-time: Deadlines must not be missed (e.g., pacemakers)

- 2. Soft real-time: Occasional delays are acceptable (e.g., multimedia systems)

### 3. Microcontroller/Microprocessor-Based

Uses microcontrollers (MCU) or System-on-Chip (SoC) with integrated CPU, memory, and peripherals.

### 4. Embedded Software (Firmware)

Runs dedicated software stored in ROM/Flash, often written in C/C++ or assembly.

### 5. Low Power Consumption

Designed to operate with minimal energy, especially in battery-powered devices.

### 6. Compact Size and Cost-Effective

Optimized for small physical size and low production cost.

### 7. High Reliability and Stability

Expected to run continuously for long periods with minimal failures.

### 8. Limited User Interface

Interfaces may include buttons, LEDs, LCDs, touchscreens, or sensors rather than keyboards and mice.

### 9. Interaction with Physical Environment

Uses sensors and actuators to monitor and control real-world processes

# Classification of Embedded Systems:

Based on Generation

Based on Complexity & Performance Requirements

# Introduction to Embedded System

## Embedded Systems - Classification based on Generation

First Generation: The early embedded systems built around 8bit microprocessors

like 8085 and Z80 and 4bit microcontrollers

Second Generation: Embedded Systems built around 16bit microprocessors and 8

or 16bit microcontrollers, following the first generation embedded systems

Third Generation: Embedded Systems built around high performance 16/32 bit Microprocessors/controllers, Application Specific Instruction set processors like Digital Signal Processors (DSPs), and Application Specific Integrated Circuits (ASICs)

Fourth Generation: Embedded Systems built around System on Chips (SoCs),

Re-configurable processors and multicore processors

# Introduction to Embedded System

## Embedded Systems - Classification based on Complexity & Performance

Small Scale: The early embedded systems built around 8bit microprocessors like 8085 and Z80 and 4bit microcontrollers

Medium Scale: Embedded Systems built around 16bit microprocessors and 8 or

16bit microcontrollers, following the first generation embedded systems

Large Scale/Complex: Embedded Systems built around high performance 16/32 bit Microprocessors/controllers, Application Specific Instruction set processors like Digital Signal Processors (DSPs), and Application Specific Integrated Circuits (ASICs)



# General-Purpose Processor (GPP)

A General-Purpose Processor (GPP) is a processing unit designed to perform a wide variety of tasks rather than a single dedicated function. It is commonly used in personal computers, laptops, servers, and workstations, where flexibility and high computational performance are required.

A General-Purpose Processor is a programmable CPU capable of executing multiple applications and handling different types of workloads such as computation, data processing, multimedia, and networking.

## Characteristics of General-Purpose Processors

- High processing speed

- Supports multitasking

- Large memory addressing capability

- Uses external RAM, ROM, and peripherals

- High power consumption

- Complex architecture

## Embedded Processor

An embedded processor is a specialized processing unit designed to perform a specific function within an embedded system.

Unlike general-purpose processors used in computers (like Intel or AMD CPUs), embedded processors are optimized for dedicated tasks, real-time operation, low power consumption, and cost efficiency.

An embedded processor is the brain of an embedded system that:

- Executes a predefined program

- Controls hardware components

- Interacts with sensors and actuators

- Works continuously for a specific application

## Features of Embedded Processors

- Dedicated application

- Real-time performance

- Low power consumption

- Small size

- High reliability

- Long operational life

## Comparison: General Processors vs Embedded Processors

| Feature            | General-Purpose Processor | Embedded Processor   |
|--------------------|---------------------------|----------------------|
| Purpose            | Multiple applications     | Single specific task |
| System Type        | Computer systems          | Embedded systems     |
| Processing Power   | Very high                 | Moderate to low      |
| Power Consumption  | High                      | Low                  |
| Size               | Large                     | Small and compact    |
| Cost               | Expensive                 | Cost-effective       |
| Operating System   | Windows, Linux, macOS     | RTOS / Bare-metal    |
| Peripherals        | External                  | Mostly integrated    |
| Real-Time Response | Not guaranteed            | Guaranteed           |
| Reliability        | Moderate                  | Very high            |

| Criteria          | General Purpose Computer                                                                                | Embedded system                                                                                          |
|-------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Contents          | It is combination of generic hardware and a general purpose OS for executing a variety of applications. | It is combination of special purpose hardware and embedded OS for executing specific set of applications |
| Operating System  | It contains general purpose operating system                                                            | It may or may not contain operating system.                                                              |
| Alterations       | Applications are alterable by the user.                                                                 | Applications are non-alterable by the user.                                                              |
| Key factor        | Performance" is key factor.                                                                             | Application specific requirements are key factors.                                                       |
| Power Consumption | More                                                                                                    | Less                                                                                                     |
| Response Time     | Not Critical                                                                                            | Critical for some applications                                                                           |

# Computer Systems

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- General-purpose computer systems



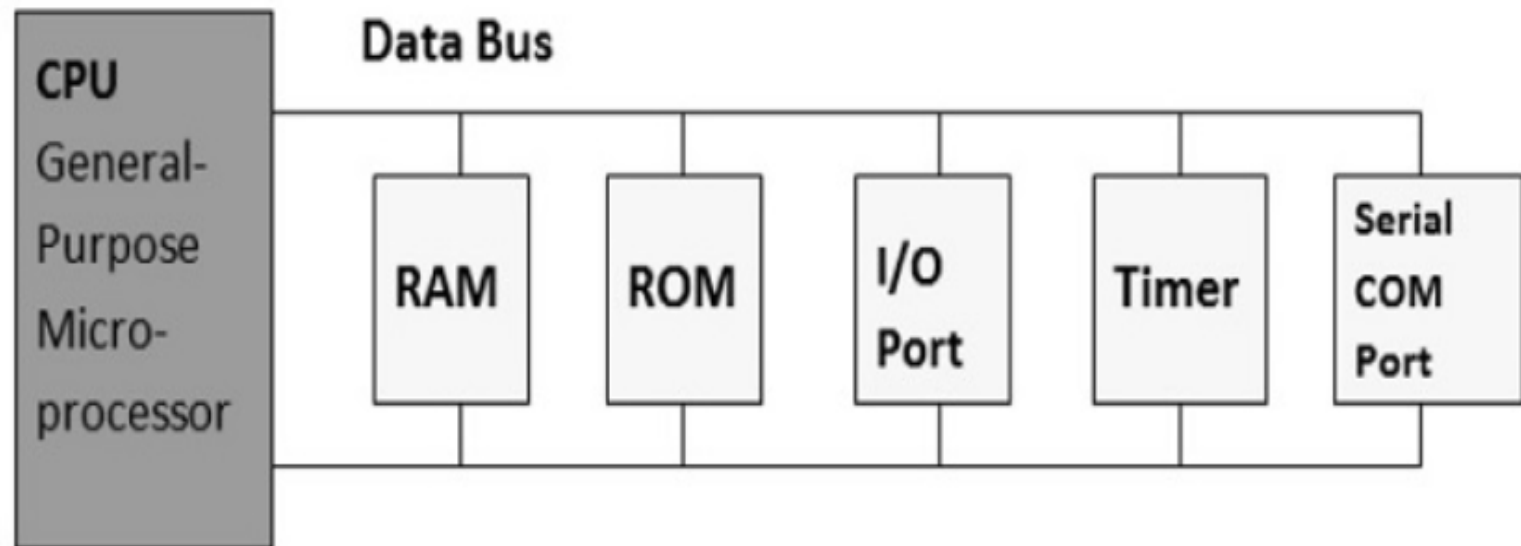
- Embedded Systems



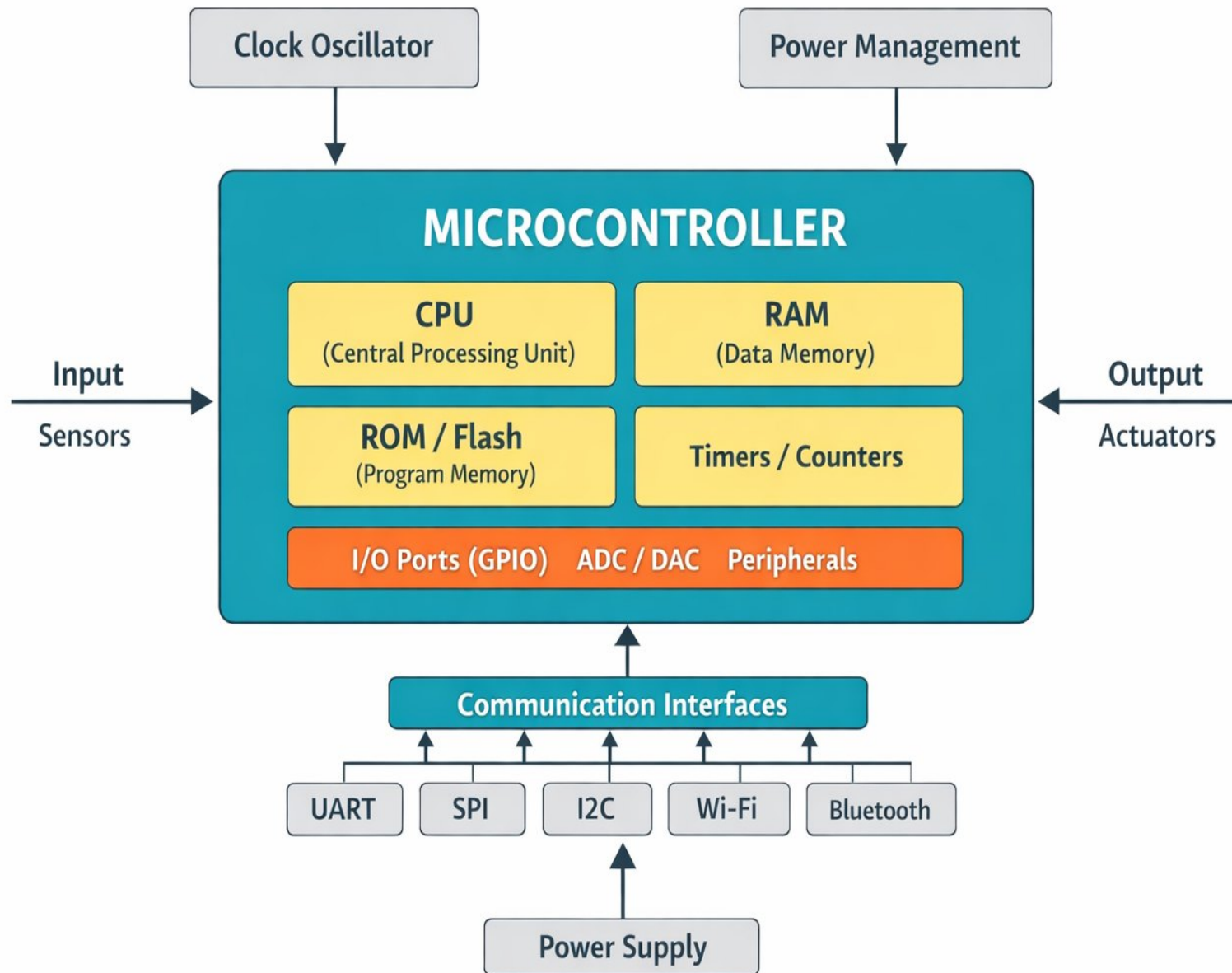
# Microprocessor

A microprocessor is a single VLSI chip having a CPU. In addition, it may also have other units such as caches, floating point processing arithmetic unit, and pipelining units that help in faster processing of instructions.

Earlier generation microprocessors' fetch-and-execute cycle was guided by a clock frequency of order of  $\sim 1$  MHz. Processors now operate at a clock frequency of 2GHz







## EMBEDDED SYSTEM AND ITS REAL TIME APPLICATIONS



- Real time systems are those systems that work within strict time constraints and provide a worst-case time estimate for critical situations. Embedded systems provide a specific function in a much larger system. When there is an embedded component in a real time system, it is known as a real time embedded system

## Real time Systems :

[www.TheEngineeringProjects.com](http://www.TheEngineeringProjects.com)

# Real-Time System

**Radar system**



**Incoming Missile**



**Command & Decision  
system**



**Weapon Firing  
system**

